

AAMOD VARMA

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science

Aug. 2023 – Expected May 2027

Major in Computer Science; Major in Mathematics

GPA: 4.0

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Linear Algebra

Private International English School

Abu Dhabi, UAE

Grade 12 CBSE - 96.6%

May. 2011 – May 2023

Awards: 1st at Hacklytics - Traversaal AI Challenge (2024); UAE National Champions, F1 in Schools (2021-22); 1st at Abu Dhabi University Programming Competition (2021); 3rd in the GCC at SOF-IMO (2020)

EXPERIENCE

Fung Group, Georgia Tech

October 2023 – Present

Undergraduate Research Assistant

Atlanta, GA

- Trained a custom word level and BPE tokenizer for dataset of chemical compositions.
- Finetuned LLMs to generate novel chemical compositions based on length and initial element.
- Generated an initial sample of 50,000 chemical compositions to be further validated.

Istiophix, F1iS

June 2021 – July 2022

Manufacturing & CFD Engineer

Abu Dhabi, UAE

- Manufactured 20+ 20mm long cars utilizing CNC and 3D printers, employing SLA and ABS printing methods.
- Conducted aerodynamic testing on 200+ prototypes of wheels, side pods, and rear pods using CFD and FEA.
- Developed a Django-based website to showcase 3D models and simulations made using Blender.

PROJECTS

TraverGo - Traversaal AI Challenge Winner | Python, Qdrant, Pandas, Git, OpenAI

Feb 2024 – Present

- Developed a 1st place-winning hotel search platform at Georgia Tech Hacklytics competing against 1000 students.
- Utilized a neural search engine to match hotels ranked by sentiment analysis conducted on over 20 reviews.
- Integrated the Ares API to provide a Q&A chatbot with real-time hotel information, enhancing user experience.

Rocket Neural Network | Python, Pytorch, NumPy, Pandas, Git

Sep 2023 – Present

- Designed a regression neural network to predict 6 aerodynamic coefficients and center of pressure values.
- Decreased mean absolute error from 0.5 to 0.01.
- Generated 20,000 bounded samples using latin hypercube sampling, decreasing discrepancy from 0.0008 to 0.0002.

Project Turtle | Python, PyTorch, Git, Django, HTML

Feb 2022 – Jan 2023

- Developed and trained a convolutional neural network using PyTorch to identify turtle tracks.
- Utilized selenium to scrape turtle footprint images from open sources automatically.
- Developed a website using Django to showcase the project.

Personal Book Management System | Python, Git, Django, PostgreSQL, HTML, API

Jan 2021 – Aug 2022

- Developed a Python GUI application using Tkinter integrated with MandaDex API to download books.
- Implemented features to automatically convert directories of chapters into the CBZ and ZIP format.
- Developed a script to convert foreign language screen captures to English using OpenCV and Pytesseract.
- Utilized Google Sheets API to fetch PostgreSQL tables and Google API to scrape cover pages for these books.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Bash, SQL, HTML/CSS, R

Tools: PyTorch, Git, Blender, NumPy, Docker, Linux, LaTeX

Essential Skills: Communication, Teamwork, Problem-solving, Adaptability, Leadership